October, 2026

Mateusz Cedro

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My research focuses on **machine learning interpretability**, with a current emphasis on **representation learning in Transformers to understand and mitigate LLM misalignment**. Across my work, I have quantified behavioural (mis)alignment in state-of-the-art LLMs [6], explained computer vision models [5], developed human-centred explanations [1,4], and interpreted multimodal ML models in medical applications [2,3]. Ultimately, I aim to **deepen our understanding of AI systems**.

Experience

Sept '24 – present	**PhD Student in Machine Learning**, University of Antwerp, Belgium, advisor: Prof. David Martens
Jan – Sept '24	**AI Research Scientist**, MI2.AI (Centre for Credible AI), Warsaw University of Technology, Poland
June – Nov '23	**Machine Learning Engineer**, Synerise, Poland
July – Dec '22	**Machine Learning Engineer**, Data Juice Lab, Poland
Feb – July '22	**Data Scientist**, Health Labs Care, Poland

Education

Sept '24 – present	**PhD Student in Machine Learning**, University of Antwerp, Belgium
Oct '21 – June '23	**MSc in Data Science**, University of Warsaw, Poland
Oct '18 – June '21	**BSc in Economics**, University of Warsaw, Poland

Publications

[6] **M. Cedro**, T. Ichmoukhamedov, S. Goethals, Y. He, J. Hinns, D. Martens. Cash or Comfort? How LLMs Value Your Inconvenience. *Communications of the ACM, 69(9), 58-66*, 2026.

[5] **M. Cedro**, M. Chlebus. Scaling Vision Models Does Not Consistently Improve Localisation-Based Explanation Quality. *Scientific Reports, 16(1), 28613*, 2026.

[4] **M. Cedro**, D. Martens. GraphXAIN: Narratives to Explain Graph Neural Networks. *World Conference on Explainable Artificial Intelligence, 91-114*, 2025.

[3] M. Dobrzycka, A. Sulewska, J. Konopinska, P. Karabowicz, A. Charkiewicz, K. Golaszewska, **M. Cedro**, P. Biecek, J. Niklinski, A. Charkiewicz, R. Charkiewicz. Machine learning-based identification of small RNA signatures in aqueous humor as a step toward precision diagnosis of glaucoma. *Annals of Medicine, 57(1), 2568119*, 2025.

[2] M. Dobrzycka, A. Sulewska, **M. Cedro**, P. Karabowicz, R. Charkiewicz, A. Charkiewicz, K. Golaszewska, P. Biecek, J. Niklinski, A. Charkiewicz, J. Konopinska. A multimodal model of aqueous humor small RNA and clinical features shows limited discrimination between primary open-angle and pseudoexfoliative glaucoma. *Scientific Reports*, 2026.

[1] **M. Cedro**, D. Martens. On the Importance and Evaluation of Narrativity in Natural Language AI Explanations. Under review, 2026.

Conference Presentations and Talks

FAIR, Research Day. Cash or Comfort? How LLMs Value Your Inconvenience (oral). May 2026.

Antwerp Center on Responsible AI. Cash or Comfort? How LLMs Value Your Inconvenience (oral). March 2026.

FAIR, Research Day. Risks of LLMs' Valuation of Human Discomfort (poster). October 2025.

World Conference on Explainable Artificial Intelligence. GraphXAIN: Narratives to Explain GNNs (oral). July 2025.

Antwerp Center on Responsible AI. Narratives to Explain AI Models (oral). February 2025.

Summer Schools and Trainings

June '26 **Machine Learning Summer School on Safety & Reliability**, focused on AI safety, understanding, and alignment of ML models. Lectures were conducted by Prof. Wojciech Samek (TU Berlin/HHI), Jan Betley (Truthful AI), and Fazl Barez (Oxford), among others.

April '26 **Technical AI Safety Training** by **BlueDot Impact**, focused on current problems and research approaches in technical AI safety, including AI alignment, mechanistic interpretability, evaluations, red-teaming, and AI control.

June '24 **Summer School on Artificial Intelligence for Biomedical Applications**, focused on machine learning modelling and interpretability with applications in medicine.

Technical Skills

Mathematics	Linear algebra, Statistics, Calculus, Probability
Programming	Python, Bash, Git, SQL
Interpretability	Representation Learning, Representational Geometry, Linear Probes, Intervening & Steering, Counterfactual & Feature Attribution explanations
ML/LLM libraries	`PyTorch`, `Transformers`, `TransformerLens`, `scikit-learn`, `NumPy`, `pandas`
Infrastructure	Linux, SSH, cloud computing, Docker
Software	Author of `xains` (Python library for natural-language XAI explanations)

Teaching

fall '26 **Python for Machine Learning** (Master's course) - lab sessions & project supervision, 120 students
spring '26 **Data Engineering** (Master's course) - lab sessions & project supervision, 50 students
fall '25 **Python for Machine Learning** (Master's course) - lab sessions & project supervision, 120 students
fall '24 **Python for Machine Learning** (Master's course) - lab sessions & project supervision, 120 students

References

Prof. Dr. David Martens
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Chair of Antwerp Center on Responsible AI, Belgium
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Prof. Dr. Przemyslaw Biecek
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